Lab 4: PID Tuning - Open-Loop Transient Response Method

1. Objective

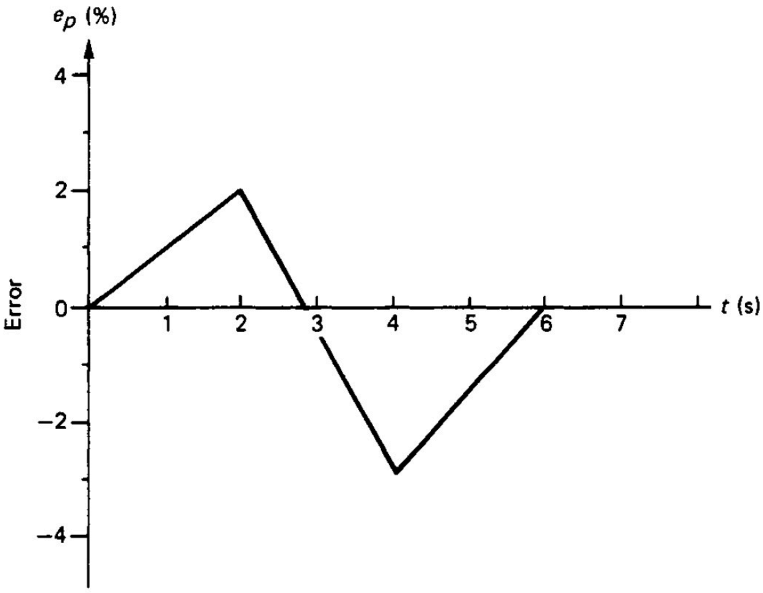
The goal of this lab is to determine the PID parameters (K_p, T_i, T_d) of a temperature control system using the **Open-Loop Transient Response Method** (also known as the **Process Reaction Curve Method**).

2. Pre-lab Task (10 marks)

Instructions: Complete these tasks before your scheduled lab session. Show your working clearly.

2.1 Proportional Controller Analysis (5 Marks)

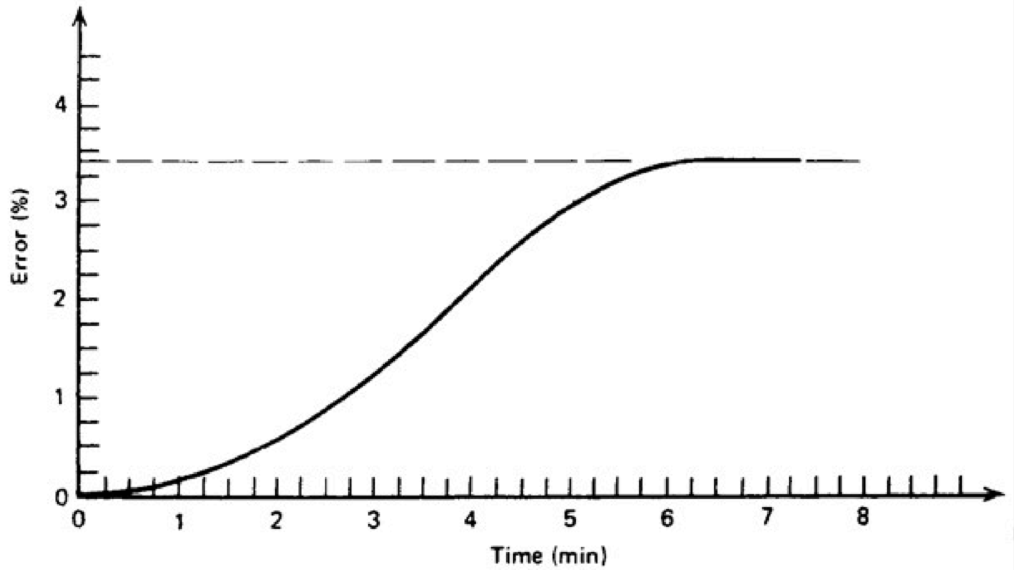
A proportional controller has a proportional gain, $K = 2.0$, and a bias, $p = 0.5$.
Determine the controller output (P) over time when the error signal shown below is inputted to the controller.



Recall the formula: $P = K \cdot e + p$

2.2 Theoretical ZN Tuning (5 Marks)

An open-loop step response of a process is given in the figure below when a step change of **7.5%** in the control variable (MV) is applied.

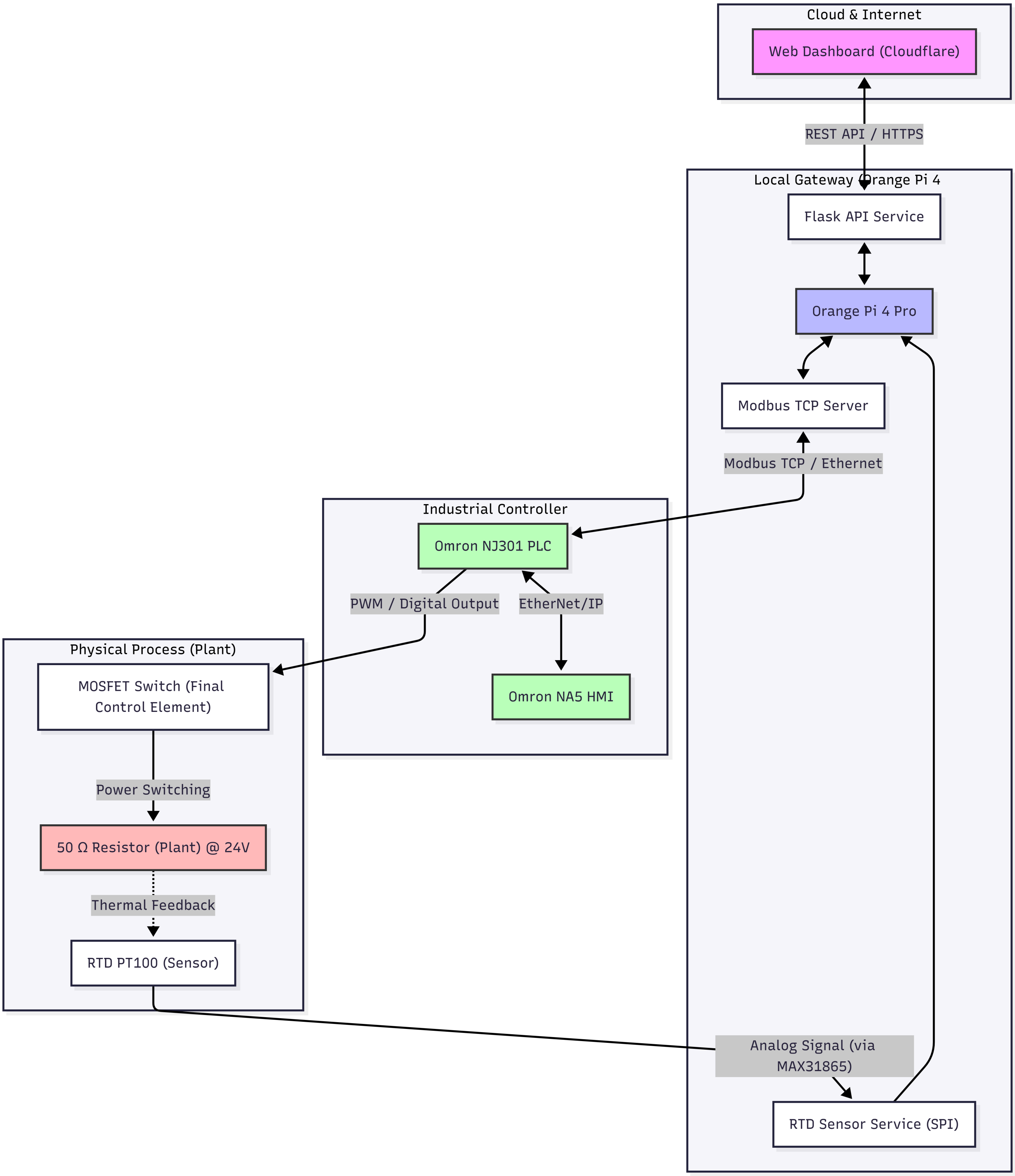


Using the Open-Loop Transient Response Method, determine the appropriate parameters for a **PID Controller** (i.e., K_p, T_i, and T_d).

3. Background

3.1 Lab Setup & Remote Architecture

Before starting the experiment, it is important to understand how your commands reach the physical hardware in the laboratory. This remote setup allows you to interact with industrial-grade equipment from anywhere.



System Components:

- **Web Dashboard**: Your user interface for real-time monitoring and control.
- **Cloud Worker**: A secure bridge that routes your dashboard commands.
- **Orange Pi Gateway**: Centrally manages data flow locally.
- **Hardware Core**: Omron NJ301 PLC, Radxa Camera, and ESP32 Smart Relay.

3.2 The Open-Loop Technique

In an open-loop test, the controller is set to **Manual Mode**. A sudden step change is applied to the Manipulated Variable (MV), and the resulting response of the Process Variable (PV) is recorded.

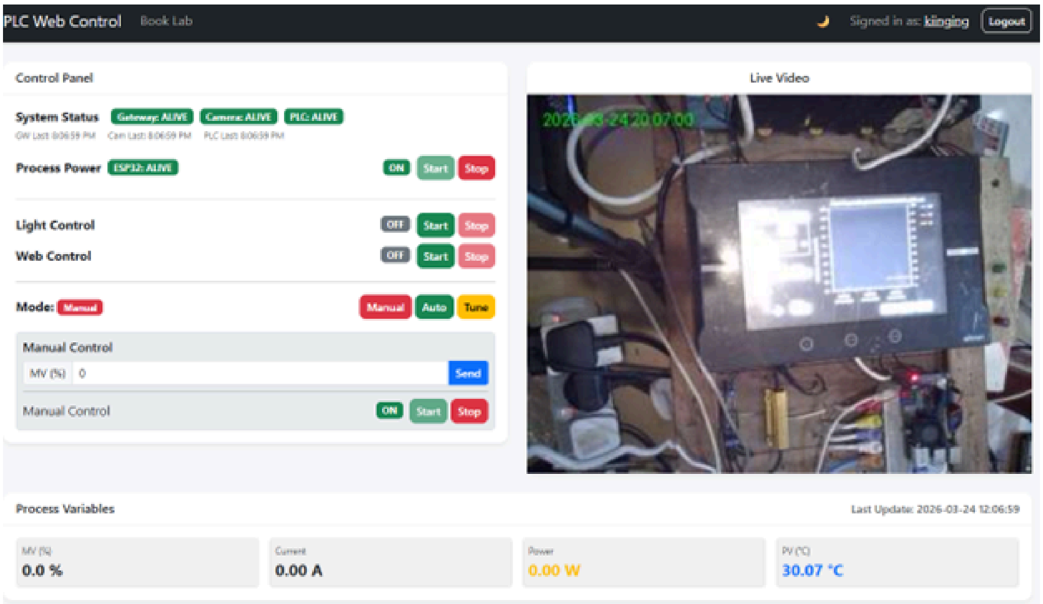
3.3 Understanding Process Gain (K)

- **0% MV** leads to a steady-state temperature of **27°C** (Ambient).
- **100% MV** leads to a steady-state temperature of **130°C**.
- **Process Gain (K)**: $K = \frac{\Delta PV}{\Delta MV}$
- Knowing the full range (**103°C**) helps in normalizing the response.

4. Measurement Tasks

4.1 System Readiness Check

1. **Dashboard Overview**: Familiarize yourself with the interface shown below.



2. **Verify Connectivity**: Look at the **System Status** badges. Both **Gateway** and **ESP32** must be **ALIVE**.
3. **Troubleshooting**: Contact: `wong.kiing.ing@curtin.edu.my` or WhatsApp `0128789001`.

4.2 Powering Up

1. **Process Power**: Click **Start** in the Process Power section.
2. **Camera**: It takes ~1 minute for the camera to come online.
3. **Light Test**: Toggle the **Light Control** while watching the video to confirm real-time control.

4.3 Thermal Experiment

1. **Web Control**: Click **Start** to enable PLC communication.
2. **Manual Initialization**: Set mode to **Manual** and MV to **40%**.
3. **Steady State**: Wait for the temperature to stabilize at 40% MV.
4. **The Step Change**: Change MV to **45%** (a 5% step) and click **Start**.
5. **Data Export**: Wait for stabilization at 45% MV, then click **CSV** to download the data.

5. Post Laboratory Task (10 marks)

5.1 Processing the recorded data (5 Marks)

1. **Time Conversion**: In Excel, convert raw time data (column D) to seconds using: `=(D_current - D_start)/(1e9)`.
2. **Smoothing**: Create a moving average (100 samples) to smooth the temperature signal: `=AVERAGE(F23:F123)`.
3. **Plotting**: Plot Temperature Response and Input Current on the same graph. Use a **Secondary Axis** for temperature to clearly see both signals.

5.2 Determination of PID control parameters (5 Marks)

1. **L and N Determination**: From your smoothed graph, draw a tangent at the inflection point to find L (Dead Time) and N (Reaction Rate/Slope).
2. **Power Step (ΔP)**: Calculate the change in heating power as a percentage of the system's maximum (**11.52 W**).
3. **Temp Change (ΔC)**: Express the temperature change as a percentage of the **103°C range**.
4. **Tuning**: Calculate K_p, T_i, T_d for P, PI, and PID modes using the Ziegler-Nichols formulas.

6. Omron NJ301 PLC Implementation

6.1 PLC Conversion

The PLC uses **Proportional Band (PB)**. Convert your calculated K_p using:

$$PB = \frac{100}{K_p}$$

6.2 Summary Checklist

- [ ] Reach steady state at 40% MV.
- [ ] Step to 45% MV.
- [ ] Calculate K_p, T_i, T_d from the captured S-curve.
- [ ] Test your PID values in **Auto Mode**.